Electrical power and energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate electrical transfer rate and energy use.

SUCCESS CRITERIA - I can...

1. Use P=IV with watts, amps and volts.
2. Use E=Pt with seconds for joules.
3. Distinguish power from total energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “power” and “voltage” to explain your model or result. Sentence starters are optional.

power: rate of energy transfer

voltage: electric potential difference; energy transferred per charge

current: rate of charge flow

01 Warm-up: bring back an earlier idea

Two equal 10 Ω resistors are connected in parallel. **Find** equivalent resistance, then compare with a series connection of the same pair.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $P = IV$; $E = P\Delta t$; $P = I^2R = V^2/R$.
Constant power for $E = P\Delta t$; last equalities apply to an ohmic resistor.

Worked reasoning: A device at 6 V drawing 2 A receives power VI=12 W. Over 5 s it receives 60 J. Power is an energy-transfer rate.

Your turn: Draw an input-energy arrow and label voltage, current, power and elapsed time with distinct units.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two motors use the same energy but one takes half the time.
Compare average power.

Sentence starter: Both … . They differ in … because … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

04 Apply the physics

A 12 V device draws a steady 0.75 A for 4 minutes.
Calculate input power and total electrical energy in joules.
Label the system model and show the time conversion.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student says a 10 W motor always uses more energy than a 5 W motor. Supply a time comparison that disproves this.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A shallow-water model uses $v=\sqrt{(gh)}$. With g=10, h=40 m and travel distance 6 km, find speed and travel time.
State a model limitation.
Label a wave sketch to support the calculation or comparison.

Word help: wave = a disturbance that can carry energy and information

Sentence starter: Under the assumption …, the evidence … supports … because … .



Time or position: choose and label

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 6 Ω resistor carries 2 A.
Find its power using two equivalent equations.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 12 V motor draws 1.5 A for 40 s and delivers 432 J of useful mechanical energy.
Find electrical input energy, average useful power and efficiency.

Sentence starter: I found … [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3